

Top 20 Epitope Candidates – HLA-A*02:01

Model: Deep FFN (MHCflurry-trained, 91.9% accuracy, 89.6% CV)

Allele: HLA-A*02:01

10 proteins scanned | 3,536 9-mers | Top 0.6% selected

Date: June 15, 2026

Top 20 Ranking (by Binding Score)

Rank	Protein	Position	Peptide	Pred	Score	P(SB)	Anchors
1	MART-1	56-64	ALMDKSLHV	SB	1.000	100.0	L-V
2	WT1	126-134	RMFPNAPYL	SB	1.000	100.0	M-L
3	WT1	187-195	SLGEQQYSV	SB	1.000	100.0	L-V
4	WT1	10-18	ALLPAVPSL	SB	1.000	100.0	L-L
5	p53	65-73	RMPEAAPPV	SB	1.000	100.0	M-V
6	M1	3-11	LLTEVETYV	SB	1.000	100.0	L-V
7	gp100	619-627	RLMKQDFS	SB	1.000	100.0	L-V
8	Tyrosinase	369-377	YMNGTMSQV	SB	1.000	100.0	M-V
9	M1	134-142	RMGAVTTEV	SB	1.000	100.0	M-V
10	gp100	178-186	MLGTHTMEV	SB	1.000	100.0	M-V
11	Tyrosinase	1-9	MLLAVLYCL	SB	1.000	100.0	L-L
12	Tyrosinase	490-498	ALLAGLVSL	SB	1.000	99.9	L-L
13	CMV pp65	40-48	RLLQTGIHV	SB	1.000	99.9	L-V
14	KRAS	28-36	FVDEYDPTI	SB	1.000	99.9	V-I
15	gp100	13-21	AVIGALLAV	SB	1.000	99.9	V-V
16	Spike RBD	87-95	KIADYNYKL	SB	0.999	99.9	I-L
17	CMV pp65	495-503	NLVPMVATV	SB	0.999	99.9	L-V
18	Tyrosinase	214-222	FLLRWEQEI	SB	0.999	99.8	L-I
19	Tyrosinase	482-490	AMVGAVLTA	SB	0.999	99.8	M-A
20	M1	58-66	GILGFVFTL	SB	0.999	99.8	I-L

Candidate Composition

By Protein

Protein	Candidates	Top Rank
Tyrosinase	5	8
WT1	3	2
gp100/PMEL	3	7
Influenza M1	3	6

Protein	Candidates	Top Rank
CMV pp65	2	13
MART-1	1	1
p53	1	5
Spike RBD	1	16
KRAS	1	14

By Anchor Residues

p2	Count	p9	Count
L	7	V	10
M	7	L	5
V	3	I	3
I	2	A	2
F	1	T	0
S	1	M	0

L-V and M-V dominate – the classic HLA-A*02:01 anchor motif.

Clinical Annotation

Rank	Peptide	Protein	Known?	Clinical Context
1	ALMDKSLHV	MART-1	Novel	Melanoma – adjacent to known epitope cluster
2	RMFPNAPYL	WT1	Validated	Leukemia/solid tumor vaccine target
3	SLGEQQYSV	WT1	Novel	WT1 zinc finger region
4	ALLPAVPSL	WT1	Novel	WT1 N-terminal domain
5	RMPEAAPPV	p53	Known region	p53 transactivation domain (mutation hotspot)
6	LLTEVETYV	M1	Novel	N-terminal M1, near immunodominant cluster
7	RLMKQDFSV	gp100	Novel	C-terminal gp100
8	YMNGTMSQV	Tyrosinase	Validated	Melanoma antigen 369-377
9	RMGAVTTEVM	M1	Novel	M1 central domain
10	MLGTHTMEV	gp100	Novel	gp100 core domain
11	MLLAVLYCL	Tyrosinase	Novel	Signal peptide – efficient ER processing

Rank	Peptide	Protein	Known?	Clinical Context
12	ALLAGLVSL	Tyrosinase	Novel	Transmembrane domain
13	RLLQTGIHV	CMV pp65	Novel	Early pp65, surface-exposed
14	FVDEYDPTI	KRAS	Novel	Effector binding region
15	AVIGALLAV	gp100	Novel	Signal peptide region
16	KIADYNYKL	Spike RBD	Novel	ACE2 binding interface
17	NLVPMVATV	CMV pp65	Validated	Immunodominant CMV epitope
18	FLLRWEQEI	Tyrosinase	Novel	Tyrosinase catalytic domain
19	AMVGAVLTA	Tyrosinase	Novel	Transmembrane- proximal
20	GILGFVFTL	M1	Validated	Gold-standard immunodominant epitope

Priority for Experimental Validation

Tier 1 – Known validated epitopes (positive controls):

GILGFVFTL, NLVPMVATV, YMNGTMSQV, RMFPNAPYL

Tier 2 – Novel candidates in validated protein regions:

ALMDKSLHV (MART-1, adjacent to known epitope), RMPEAAPPV (p53 mutation hotspot), LLTEVETYV (M1 N-terminus), KIADYNYKL (Spike RBD ACE2 interface)

Tier 3 – Novel candidates in less-characterized regions:

SLGEQQYSV (WT1), FVDEYDPTI (KRAS effector domain), RLLQTGIHV (CMV pp65 early region)

Output Files

File	Contents
data/top20_epitope_candidates.csv	Machine-readable table
top20_epitope_candidates.md	This report
data/protein_epitope_scan_extended.csv	Full 2,764 peptide scan

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